

Project Report Format

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1. Introduction

Employee management is one of the most important functions in every organization. Managing employee onboarding and offboarding manually often leads to delays, paperwork, communication gaps, and human errors. Organizations require an automated system that can efficiently manage employee records, approvals, notifications, and departmental tasks.

The **Employee Management System** is a ServiceNow-based application developed using **App Engine Studio** to automate employee lifecycle processes. The application provides a centralized platform for managing employee information, onboarding requests, offboarding requests, approval workflows, email notifications, and task assignments for HR, IT, and Security teams. By automating these processes, the system improves efficiency, reduces manual effort, and ensures a smooth employee experience.

2. Project Overview

The **Employee Management System** is a low-code application developed on the **ServiceNow Platform** using **App Engine Studio** and **Flow Designer**. The application manages the complete employee onboarding and offboarding process through automated workflows.

The system allows employees or HR administrators to create onboarding and offboarding requests. Once a request is submitted, it is automatically routed to the appropriate manager for approval. Based on the approval decision, the system performs predefined actions such as creating HR, IT, and Security tasks, sending email notifications, updating request status, and maintaining employee records.

The project also implements **Access Control Lists (ACLs)** to ensure that only authorized users can access and modify application data. The automation provided by the system minimizes manual work, improves process transparency, and enables better coordination between departments.

3. Purpose of the Project

The primary purpose of this project is to automate the employee onboarding and offboarding process using the ServiceNow platform. The project aims to eliminate manual processing, reduce administrative workload, and improve the efficiency of HR operations.

The objectives of the project include:

- Automate employee onboarding and offboarding workflows.
- Maintain centralized employee records.
- Implement manager approval workflows.
- Automatically generate HR, IT, and Security tasks.
- Send automated email notifications during each process.
- Improve process tracking and transparency.
- Secure application data using Role-Based Access Control (ACLs).
- Reduce manual errors and increase organizational productivity.

By implementing this solution, organizations can manage the employee lifecycle more efficiently while providing a faster, secure, and standardized process for all stakeholders.

2.IDEATION PHASE

2.2 Empathize & Discover

Date	28 June 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Off boarding System
Maximum Marks	4 Marks

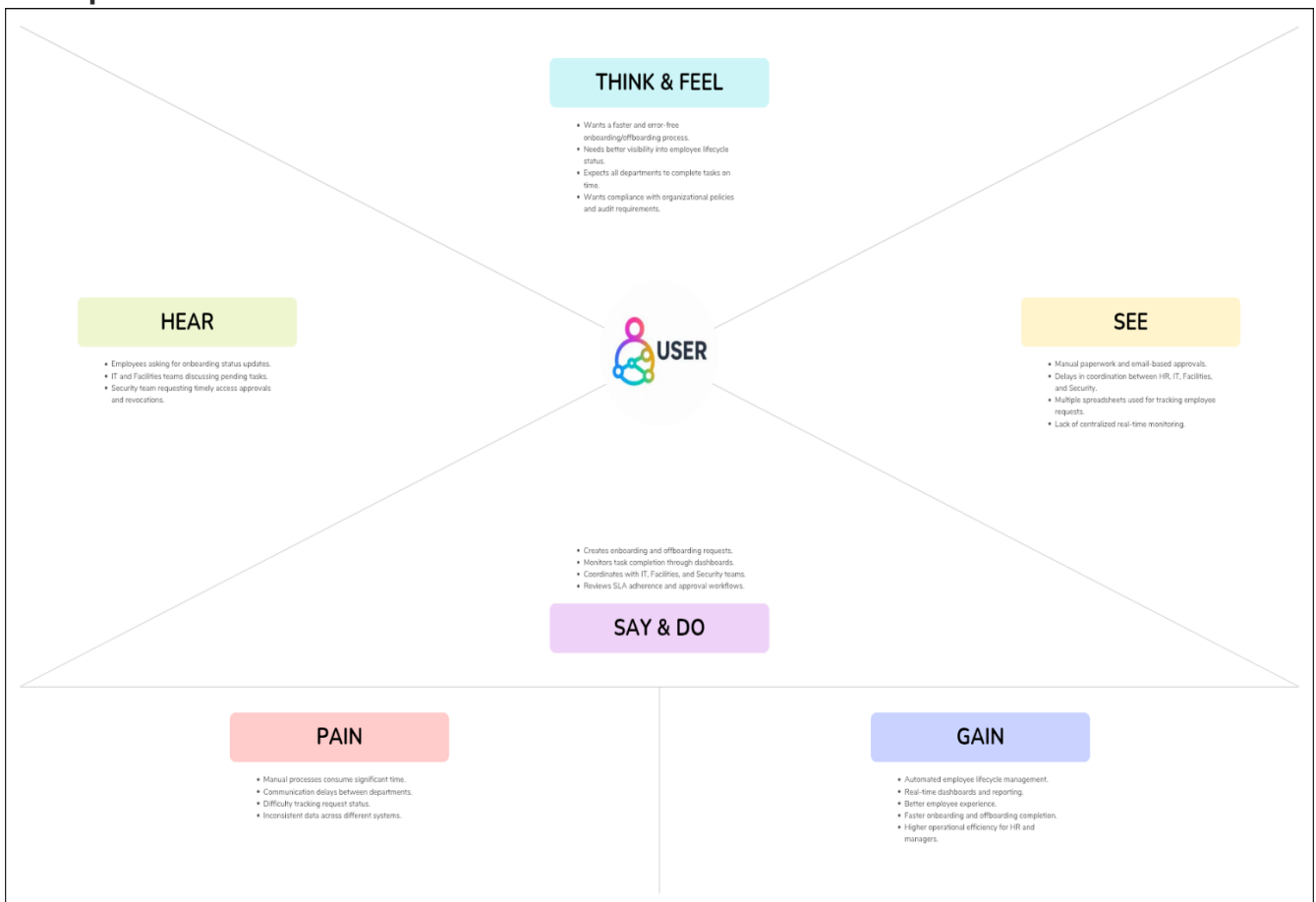
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



Ideation Phase

Define the Problem Statements

Date	28 June 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Off boarding System
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

Example:

Problem Statement(PS)	PS-1	PS-2
I AM(Customer)	HR Manager	Security Team Member
I'm Trying To	Manage employee onboarding and offboarding quickly and accurately	Complete employee access provisioning and deactivation on time
BUT	The process involves many manual approvals, paperwork, and coordination across departments.	I receive requests late and have limited visibility into task status
BECAUSE	There is no centralized automated workflow to manage employee lifecycle tasks.	Requests and approvals are not automated or properly tracked.
PWHICH MAKES ME FEEL	Frustrated due to delays, increased workload, and the risk of errors.	Stressed, inefficient, and concerned about security and compliance.

Ideation Phase

Brainstorm & Idea Prioritization Template

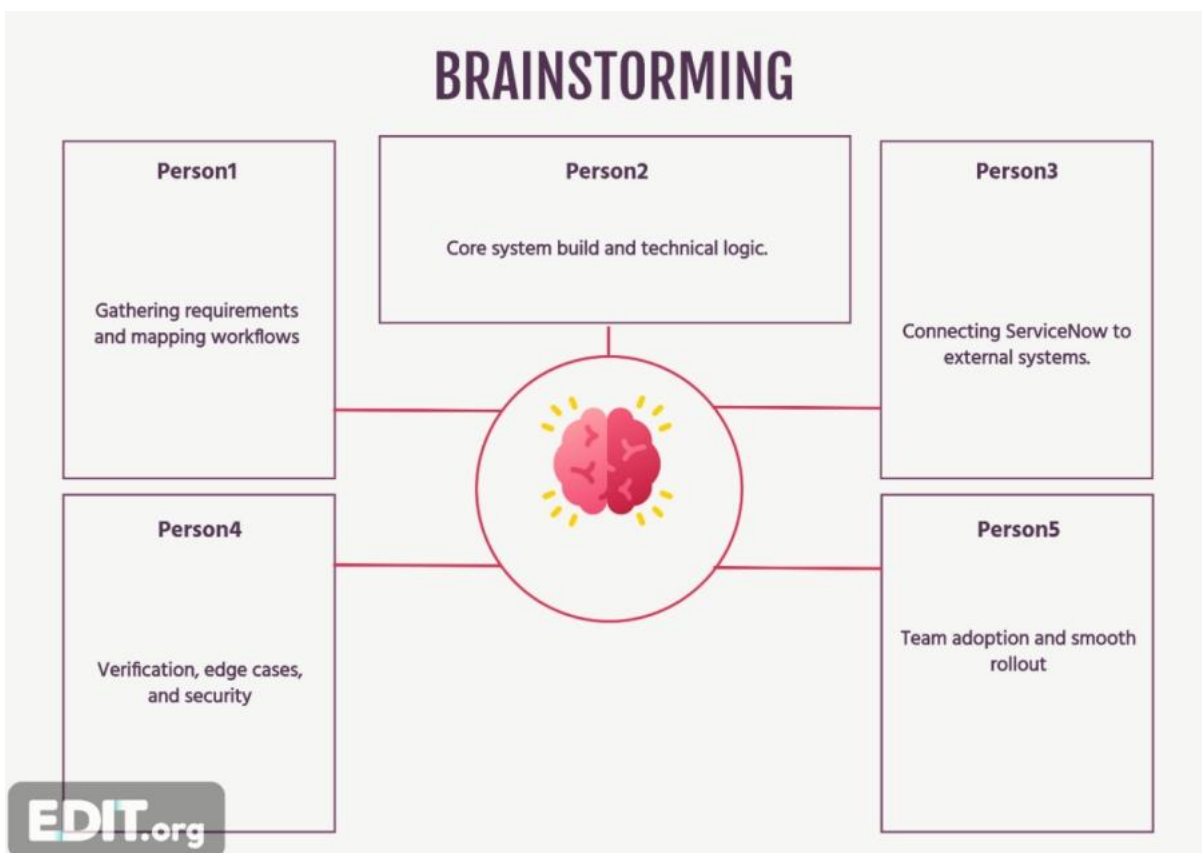
Date	6 July 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Off boarding System
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step 2: Brainstorming, Idea Listing and Grouping

S.No	Idea Description	Category
1	Automate employee onboarding request process	Workflow Automation
2	Automate employee offboarding process	Workflow Automation
3	Create Service Catalog items for onboarding and offboarding	Service Catalog
4	Generate automatic tasks for HR, IT, Facilities and Security	Task Automation
5	Configure approval workflows using Flow Designer	Approval Management
6	Implement role-based ACLs for secure access	Security
7	Configure SLA monitoring for task completion	SLA Management
8	Send automatic email notifications	Notification System
9	Design HR dashboards and reports	Dashboard & Reporting
10	Track employee lifecycle status in real time	Employee Tracking
11	Test complete onboarding/offboarding workflow	Testing
12	Improve compliance using audit trails	Compliance

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	06 July 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Off boarding System
Maximum Marks	4 Marks

Solution Requirements - Employee Management System
Functional Requirements

FR No	Functional Requirement	Description
FR-1	Employee Management	Create and maintain employee records.
FR-2	Onboarding Process	Submit onboarding request, manager approval, create HR/IT/Security tasks.
FR-3	Offboarding Process	Submit offboarding request, manager approval, create exit tasks.
FR-4	Workflow Automation	Automate approvals, email notifications and record updates using Flow Designer.
FR-5	Access Control	Allow only authorized users to create, read and update records using ACLs.

Non-Functional Requirements

NFR No	Requirement	Description
NFR-1	Usability	Simple and user-friendly forms in App Engine Studio.
NFR-2	Security	Role-based Access Control (ACL) protects application data.
NFR-3	Reliability	Flows execute consistently with approvals and notifications.
NFR-4	Performance	Automation reduces manual effort and processing time.
NFR-5	Availability	Application is available on the ServiceNow cloud platform.
NFR-6	Scalability	Can be extended with additional HR processes and modules.

Project Design Phase-II

Data Flow Diagram & User Stories

Date	06 July 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Off boarding System
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Employee	Employee Registration	USN-1	As an employee, I can create a new employee record with personal and job details.	Employee record is successfully created.	High	Sprint-1
Employee	Onboarding Request	USN-2	As an employee, I can submit an onboarding request.	Onboarding request is submitted successfully.	High	Sprint-1
Manager	Approval	USN-3	As a manager, I can approve or reject onboarding requests.	Manager can approve/reject and status updates correctly.	High	Sprint-1
HR	Employee Onboarding	USN-4	As an HR user, I can receive and complete onboarding tasks.	HR task is automatically created after approval.	High	Sprint-1
IT	IT Provisioning	USN-5	As an IT user, I can receive tasks to create user accounts	IT task is created automatically.	Medium	Sprint-1

			and assign assets.			
Employee	Offboarding Request	USN-6	As an employee, I can submit an offboarding request.	Offboarding request is submitted successfully.	High	Sprint-2
Manager	Offboarding Approval	USN-7	As a manager, I can approve or reject offboarding requests.	Status updates based on manager decision.	High	Sprint-2
HR	Exit Process	USN-8	As an HR user, I can process employee exit formalities.	HR offboarding task is created automatically.	Medium	Sprint-2
IT	Access Revocation	USN-9	As an IT user, I can disable employee accounts and collect assets.	IT offboarding task is completed successfully.	Medium	Sprint-2
Security	Building Access	USN-10	As a security user, I can deactivate employee building access.	Security task is created and completed successfully.	Medium	Sprint-2
Administrator	Access Control	USN-11	As an administrator, I can manage roles and access permissions for users.	Authorized users can access only permitted records.	High	Sprint-2
Administrator	Notifications	USN-12	As an administrator, I can configure email notifications for approvals and requests.	Email notifications are sent successfully.	Medium	Sprint-2

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	06 July 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Offboarding System
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2

Technology Stack - Employee Management System

Component	Description	Technology
User Interface	Application forms and lists	ServiceNow App Engine Studio
Application Logic	Workflow automation	Flow Designer
Approval Engine	Manager approvals	Ask for Approval Action
Database	Employee, Onboarding, Offboarding tables	ServiceNow Custom Tables
Notifications	Email alerts	ServiceNow Notification/Send Email
Security	Access Control	Roles and ACLs
Platform	Deployment	ServiceNow Personal Developer Instance (Cloud)

Project Design Phase

Problem – Solution Fit Template

Date	06 July 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Off boarding System
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ Understand the existing situation in order to improve it for your target group.

Section	Description
1. Customer Segment(s)	Employees, HR Team, Managers
2. Jobs-to-be-done	Employee Onboarding, Employee Offboarding, Approval Process
3. Triggers	New Employee Joining, Employee Resignation
4. Emotions (Before / After)	Before: Manual Process, Delays. After: Fast, Automated and Transparent Process
5. Available Solutions	Manual Forms, Email Process, ServiceNow Automation
6. Customer Constraints	Manual Work, Delayed Approvals, Tracking Issues
7. Behaviour	HR verifies requests, Manager approves/rejects requests, IT completes assigned tasks
8. Channels	Online: ServiceNow Platform, Email Notifications. Offline: HR Team Communication
9. Problem Root Cause	Manual Process, No Centralized Tracking, Human Errors
10. Your Solution	Developed an Employee Management System using ServiceNow App Engine Studio. Automated employee onboarding and offboarding with Flow Designer, Manager Approval, Email Notifications, HR/IT/Security Tasks, and Access Control (ACLs).

Project Design Phase

Proposed Solution Template

Date	06 July 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Off boarding System
Maximum Marks	2 Marks

Proposed Solution Template

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Organizations face delays and manual effort in employee onboarding and offboarding. Employee records, approvals, task assignments, and notifications are handled manually, leading to errors, lack of transparency, and inefficient tracking.
2.	Idea / Solution description	Develop an Employee Management System using ServiceNow App Engine Studio . The application automates employee onboarding and offboarding through custom tables, Flow Designer workflows, manager approvals, email notifications, HR/IT/Security task creation, and Access Control Lists (ACLs).
3.	Novelty / Uniqueness	Provides a centralized platform that integrates employee management, workflow automation, approvals, notifications, and security within a single ServiceNow application, reducing manual intervention.
4.	Social Impact / Customer Satisfaction	Improves employee experience with faster onboarding and offboarding, reduces HR workload, increases transparency, and

		ensures timely completion of organizational processes.
5.	Business Model (Revenue Model)	Can be deployed as an enterprise HR workflow solution for organizations using ServiceNow. Revenue can be generated through software licensing, implementation services, customization, and annual maintenance.
6.	Scalability of the Solution	The solution can be extended to include leave management, attendance, payroll integration, asset management, employee self-service, and additional HR workflows wit

Project Design Phase

Solution Architecture

Date	06 July 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Off boarding System
Maximum Marks	4 Marks

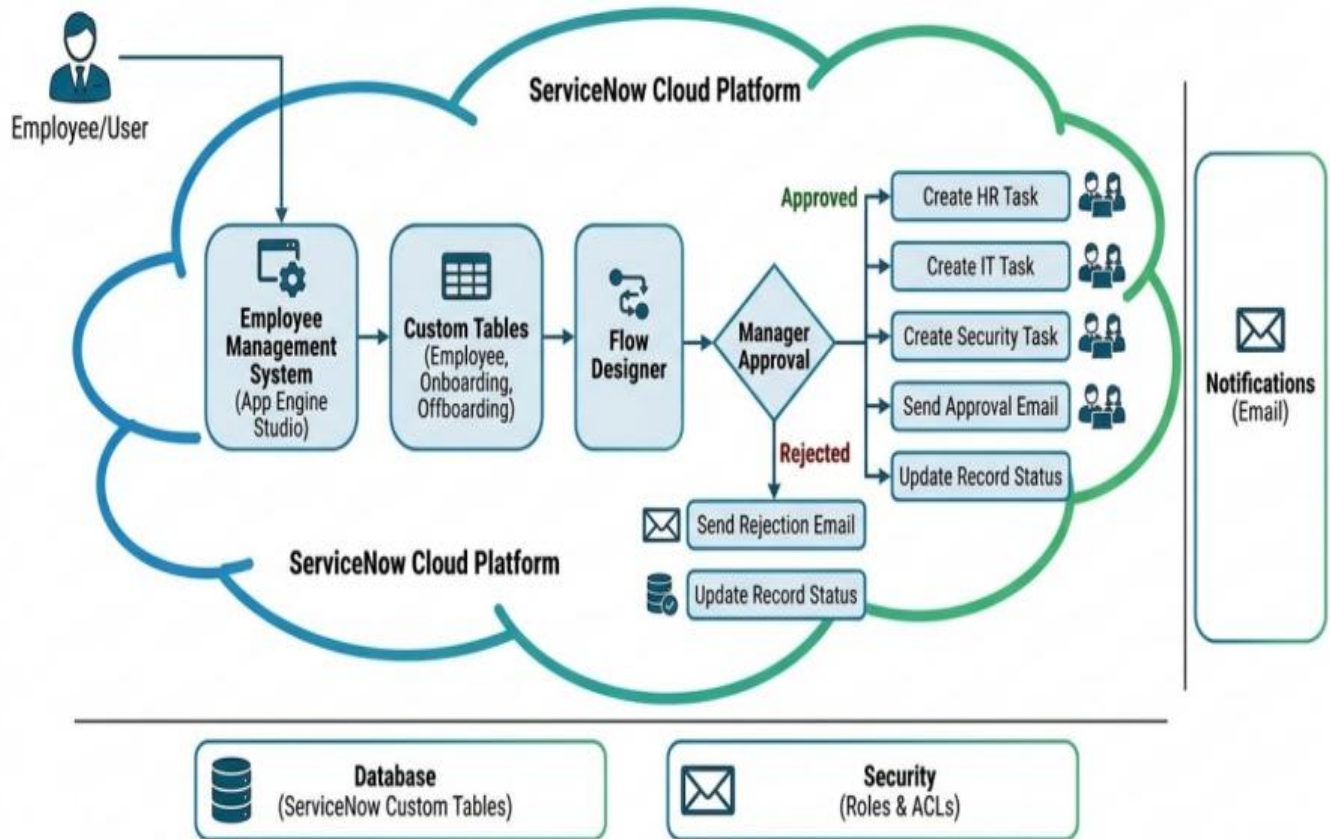
Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

Employee Management System – Onboarding & Offboarding Automation



Sprint 1

Employee Management (Epic 1)

User Story	Story Points
------------	--------------

Create Employee Table (USN-1)	2
-------------------------------	---

Create Employee Form (USN-2)	2
------------------------------	---

Create Onboarding Table (USN-3)	3
---------------------------------	---

Onboarding Process (Epic 2)

User Story	Story Points
------------	--------------

Build Onboarding Flow (USN-4)	5
-------------------------------	---

Configure Approval Process (USN-5)	3
------------------------------------	---

Configure Email Notification (USN-6)	2
--------------------------------------	---

Total Story Points Sprint-1

$2 + 2 + 3 + 5 + 3 + 2 = 17$

Sprint 2

Offboarding Process (Epic 3)

User Story	Story Points
------------	--------------

Create Offboarding Table (USN-7)	3
----------------------------------	---

Build Offboarding Flow (USN-8) 5
 Create HR/IT/Security Tasks (USN-9) 5
 Security (Epic 4)
 User Story Story Points
 Configure Access Control (USN-10) 3
 Testing & Validation (USN-11) 2
 Total Story Points Sprint-2
 $3 + 5 + 5 + 3 + 2 = 18$

Total Story Points
 Sprint-1 = 17
 Sprint-2 = 18
 Total Story Points = 35
 Number of Sprints = 2
 Velocity = $35 / 2 = 17.5 \approx 18$ Story Points per Sprint

Project Planning Phase
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	06 July 2026
Team ID	SWTID-2026-2221
Project Name	Automated Employee Onboarding & Offboarding System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Employee Management	USN-1	Create Employee Table	2	High	Team
Sprint-1	Employee Management	USN-2	Create Employee Form	2	High	Team
Sprint-1	Employee Management	USN-3	Create Onboarding Table	3	High	Team

Sprint-1	Workflow Automation	USN-4	Develop Onboarding Flow	5	High	Team
Sprint-1	Workflow Automation	USN-5	Configure Manager Approval	3	High	Team
Sprint-1	Notification	USN-6	Configure Email Notification	2	Medium	Team
Sprint-2	Employee Offboarding	USN-7	Create Offboarding Table	3	High	Team
Sprint-2	Employee Offboarding	USN-8	Develop Offboarding Flow	5	High	Team
Sprint-2	Task Automation	USN-9	Create HR, IT & Security Tasks	5	High	Team
Sprint-2	Security	USN-10	Configure Access Control (ACL)	3	Medium	Team
Sprint-2	Testing	USN-11	Test and Validate Complete Application	2	High	Team

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date
Sprint-1	17	6 Days	01 Feb 2025	06 Feb 2025	17	06 Feb 2025
Sprint-2	18	6 Days	08 Feb 2025	13 Feb 2025	18	13 Feb 2025

Acceptance Testing

UAT Execution & Report Submission

Date	03 November 2022
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - xxx
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the [ProductName] project at the time of the release to User Acceptance Testing (UAT).

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	10	4	2	3	20
Duplicate	1	0	3	0	4

External	2	3	0	1	6
Fixed	11	2	4	20	37
Not Reproduced	0	0	1	0	1
Skipped	0	0	1	1	2
Won't Fix	0	5	2	1	8

Totals 24 14 13 26 77

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and untested

Section	Total Cases	Not Tested	Fail	Pass
Print Engine	7	0	0	7
Client Application	51	0	0	51
Security	2	0	0	2
Outsource Shipping	3	0	0	3
Exception Reporting	9	0	0	9
Final Report Output	4	0	0	4

7. RESULTS

servicenow All Favorites History Workspaces Admin ServiceNow Search

app e

FAVORITES
No Results

ALL RESULTS

- App Engine
 - Creator Studio
 - App Engine Studio**
 - ServiceNow Studio
 - Table Builder
 - Configuration
 - Properties

where with Build Agent

studio, Cursor, Claude Code and others.

type with full governance

existing applications

documentations

GO FURTHER

Power your workflow applications

https://dev284127.service-now.com/now/appenginestudio

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Home

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App HomeEmployee Managemen...

Employee Management System

Source controlSubmit

All (25)Data (5)Experience (4)Logic and automation (8)Security (8)

Search data

DataAdd

Sort by: Newest

Tables that store and gather related data

Task Table 2

Table

Table is a way to store your data an...

Last edited 2026-07-03 01:27:59

PREVIEW

Project Table

Table

Table is a way to store your data an...

Last edited 2026-07-03 01:26:50

PREVIEW

Offboarding Request

Table

Table is a way to store your data an...

Last edited 2026-07-02 01:45:47

PREVIEW

Onboarding Request

Table

Table is a way to store your data an...

Last edited 2026-06-30 23:14:20

PREVIEW

Employee

Table

Table is a way to store your data an...

Last edited 2026-06-30 21:16:33

PREVIEW

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App HomeEmployee Managemen...Data Table And FormsEmployee

Employee

DataFormsFlows

Table fieldsAdd new field

More actions

	Column label *	Column name *	Type *	Reference	Max length	Default va
	Updates	sys_mod_count	Integer			
	Updated by	sys_updated_by	String		40	
	Updated	sys_updated_on	Date/Time			
	Status	status	Choice3 Choices			
	Manager	manager	Reference	User		
	Joining Date	joining_date	Date			
	Employee Name	employee_name	String		40	

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Onboarding Request

DataFormsFlows

Policies and rulesPreviewSave

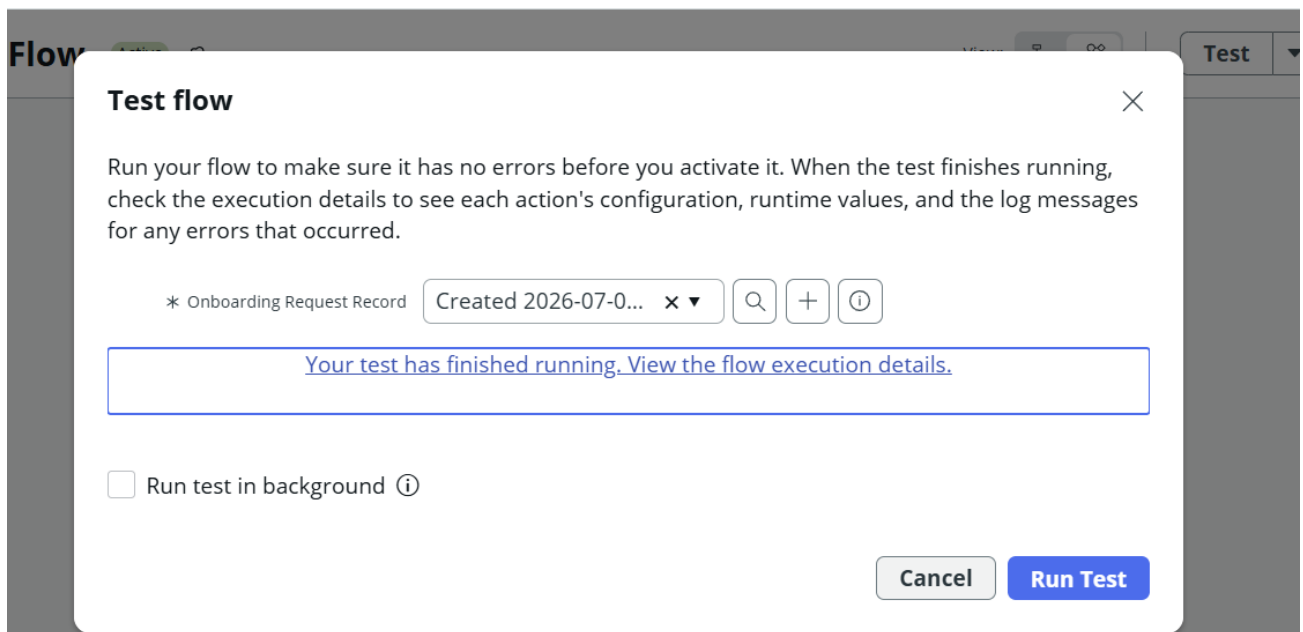
Table fieldsAdd new field

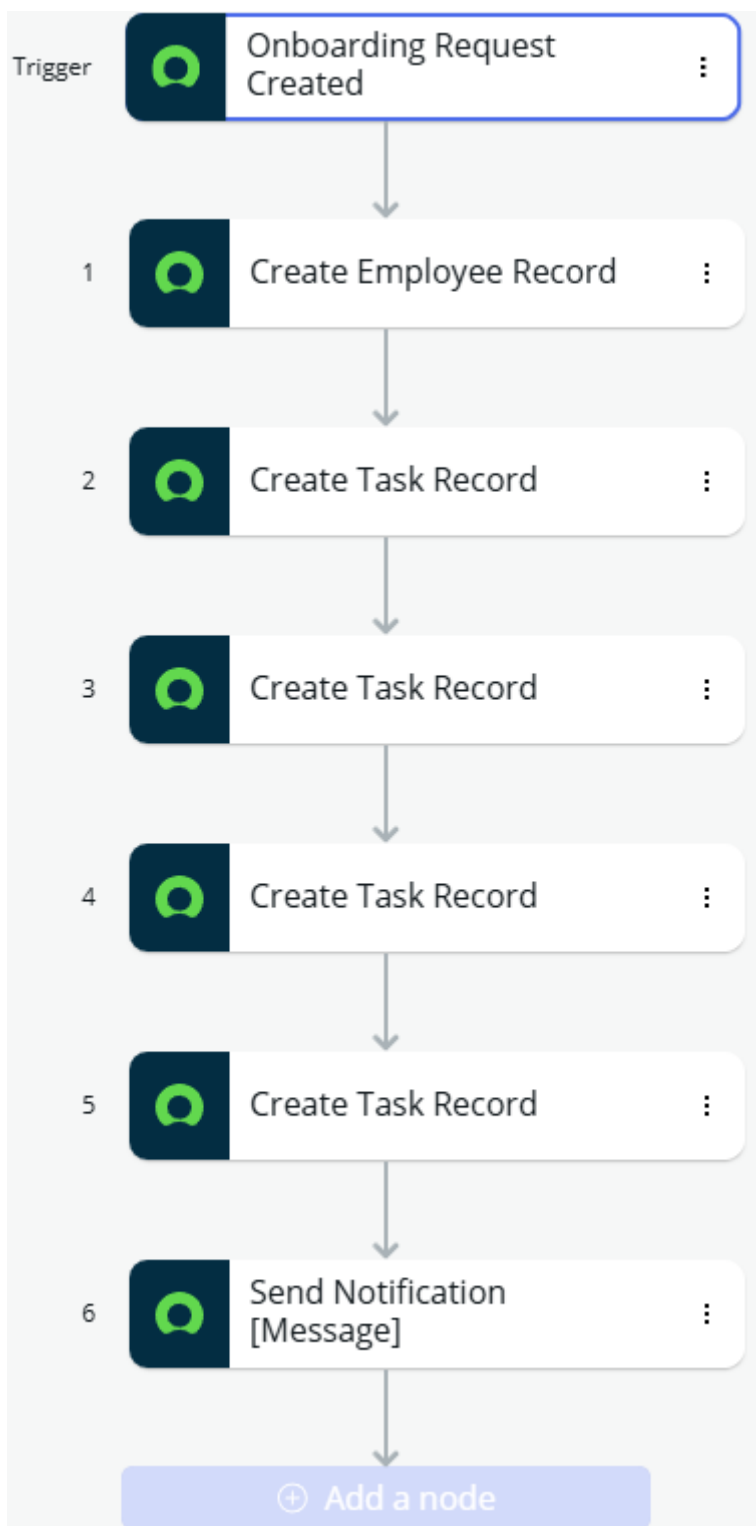
More actions

Filter fieldsFilter options

Column label *	Column name *	Type *	Reference	Max length	Default value	Display	Updated
Updates	sys_mod_count	Integer					2026-06-30 23:14:22
Updated by	sys_updated_by	String		40			2026-06-30 23:14:22
Updated	sys_updated_on	Date/Time					2026-06-30 23:14:22
Status	comments	Choice5 Choices					2026-07-01 11:06:24
Requested By	requested_by	Reference	User				2026-07-01 09:39:37
Request Number	request_number	String		40			2026-07-01 09:39:38
Joining Date	joining_date	Date					2026-07-01 09:39:37
Employee Name	employee_name	String		100			2026-07-02 01:13:44

Add new field





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App HomeEmployee Managemen...Data Table And FormsOnboarding Request4057e5efbc310310bd0...Executions

EXECUTION DETAILSEmployee Onboarding FlowTest Run - CompletedOpen flowOpen context record

Show Action Details

FLOW STATISTICSRun as: System AdministratorOpen flow logsCompleted2026-07-06 10:32:18180ms

TRIGGEROnboarding Request CreatedOpen current record

ACTIONSExpand or collapse Action Details

1	Create Record	Core Action	Completed	2026-07-06 10:32:18	30ms
2	Create Record	Core Action	Completed	2026-07-06 10:32:18	77ms
3	Create Record	Core Action	Completed	2026-07-06 10:32:18	24ms
4	Create Record	Core Action	Completed	2026-07-06 10:32:18	19ms

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App HomeEmployee Managemen...Data Table And FormsOffboarding Request

Offboarding RequestDataFormsFlowsPolicies and rulesPreviewSave

Table fieldsAdd new fieldMore actionsFilter fieldsFilter options

Column label *	Column name *	Type *	Reference	Max length	Default value	Display	Updated
Updates	sys_mod_count	Integer				☐	2026-07-02 01:45:47
Updated by	sys_updated_by	String		40		☐	2026-07-02 01:45:47
Updated	sys_updated_on	Date/Time				☐	2026-07-02 01:45:47
Status	status	Choice3 Choices				☐	2026-07-02 01:51:02
Reason	reason	Choice4 Choices				☐	2026-07-02 01:51:03
Last Working Date	last_working_date	Date				☐	2026-07-02 01:51:04
Employee	employee	Reference	Employee			☐	2026-07-02 01:51:04
Department	department	String		40		☐	2026-07-02 01:51:04

Add new field

Test flow

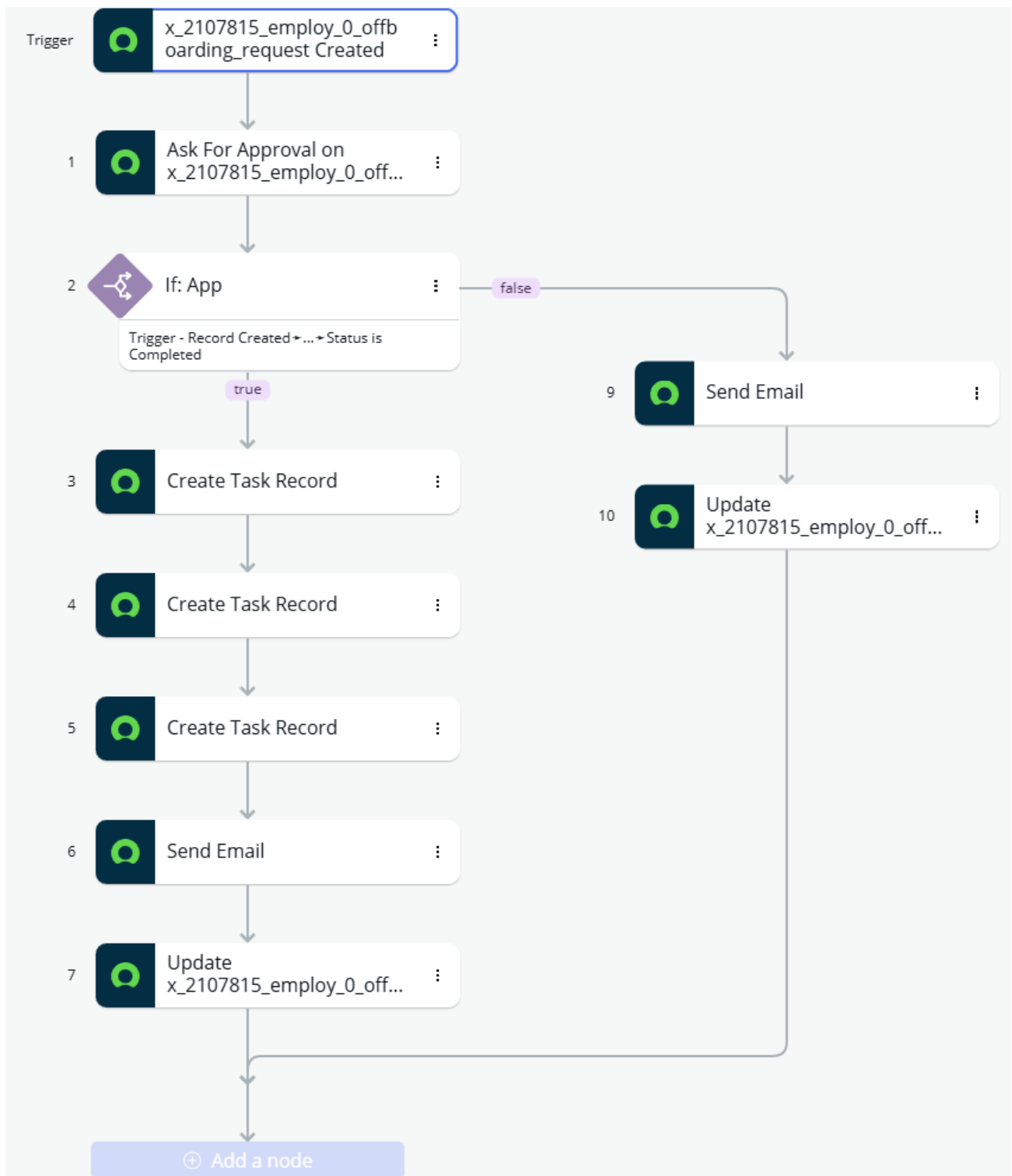
Run your flow to make sure it has no errors before you activate it. When the test finishes running, check the execution details to see each action's configuration, runtime values, and the log messages for any errors that occurred.

* RecordCreated 2026-07-0...x

Your test has finished running. View the flow execution details.

☐ Run test in background

CancelRun Test



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App Home

Employee Managemen...

Data Table And Forms

Offboarding Request

F028a563ba710310a81... x

Executions

EXECUTION DETAILS

Offboarding Request

Test Run - Completed

Open flow

Open context record

Show Action Details

State

Start time

FLOW STATISTICS

Run as: System Administrator

Open flow logs

Completed

2026-07-06 10:35:54

130ms

TRIGGER

x_2107815_employ_0_offboarding_request Created

Open current record

ACTIONS

1

Ask For Approval

Core Action

Completed

2026-07-06 10:35:54

22ms

2

If App

Flow Logic

Evaluated - True

2026-07-06 10:35:54

108ms

3

Create Record

Core Action

Completed

2026-07-06 10:35:54

24ms

4

Create Record

Core Action

Completed

2026-07-06 10:35:54

24ms

servicenowApp Engine Studio

Home

My Apps

App Home

Employee Managemen...

Employee Management System

All (26)Data (5)Experience (5)Logic and automation (8)Security (8)

Employee Onboarding Fl...

Flow - record

This flow automates the employee o.

Message

Email

notifying

Email

notifying

Email

servicenow

AllFavoritesHistoryWorkspacesAdmin

Application Menu - Project Table

Search

Application MenuProject Table

CategoryCustom Applications

The text that appears in a tooltip when a user points to this application menu

Hint

Description

UpdateDelete

ModulesOrderSearch

Actions on selected rows...New

Application menu = Project Table

Title	Table	Active	Filter	Order	Link type	Device type	Roles	Updated
Project Tables	Project Table [x_2107815_employ_0_u_project_table]	true			List of Records		x_2107815_employ_0.project_table_user	2026-07-03 01:26:51

1 to 1 of 1

No templates are availableCreate A New One?

servicenow

AllFavoritesHistoryWorkspacesAdmin

Access Controls

Search

Access ControlsUpdated

Search

Actions on selected rows...

All

Name	Decision Type	Operation	Type	Active	Updated by	Updated
Search	Search	Search	Search	Search	Search	Search
x_2107815_employ_0_u_task_table_2	Allow If	read	record	true	admin	2026-07-03 01:28:00
x_2107815_employ_0_u_task_table_2	Allow If	delete	record	true	admin	2026-07-03 01:28:00
x_2107815_employ_0_u_task_table_2	Allow If	create	record	true	admin	2026-07-03 01:28:00
x_2107815_employ_0_u_task_table_2	Allow If	write	record	true	admin	2026-07-03 01:28:00
x_2107815_employ_0_u_project_table	Allow If	delete	record	true	admin	2026-07-03 01:26:53
x_2107815_employ_0_u_project_table	Allow If	read	record	true	admin	2026-07-03 01:26:52
x_2107815_employ_0_u_project_table	Allow If	create	record	true	admin	2026-07-03 01:26:52
x_2107815_employ_0_u_project_table	Allow If	write	record	true	admin	2026-07-03 01:26:52
x_2107815_employ_0_onboarding_request	Allow If	create	record	true	admin	2026-07-02 05:04:03
x_2107815_employ_0_onboarding_request	Allow If	read	record	true	admin	2026-07-02 05:01:21

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8. ADVANTAGES & DISADVANTAGES

Advantages of the Employee Management System

- 1. Automates Employee Lifecycle**
 - Automates onboarding and offboarding processes, reducing manual work and paperwork.
- 2. Centralized Employee Data**
 - Stores employee, onboarding, and offboarding information in a centralized ServiceNow application for easy management.
- 3. Workflow Automation**
 - Uses Flow Designer to automate approvals, task creation, email notifications, and status updates.
- 4. Faster Approval Process**
 - Manager approvals are processed electronically, reducing delays and improving efficiency.
- 5. Automatic Task Creation**
 - HR, IT, and Security tasks are automatically generated after approval, ensuring that all departments complete their responsibilities on time.
- 6. Improved Communication**
 - Email notifications keep employees and stakeholders informed about request status and approvals.
- 7. Enhanced Security**
 - Access Control Lists (ACLs) ensure that only authorized users can access or modify application data.
- 8. Reduced Human Errors**
 - Automation minimizes data entry mistakes and improves process accuracy.
- 9. Easy Tracking**
 - Employees and administrators can monitor the status of onboarding and offboarding requests throughout the process.
- 10. Scalable Solution**
 - The application can be extended with additional HR modules such as leave management, payroll integration, attendance tracking, and asset management.

Disadvantages of the Employee Management System

- 1. ServiceNow Dependency**
 - The application depends entirely on the ServiceNow platform and cannot function independently.
- 2. Initial Configuration Time**
 - Setting up tables, workflows, approvals, notifications, and security requires initial development effort.
- 3. Training Requirement**
 - HR staff and end users may require training to use the application effectively.
- 4. Customization Complexity**
 - Adding advanced features or integrating with external systems may require additional ServiceNow development skills.
- 5. Internet Requirement**
 - As a cloud-based application, a stable internet connection is required to access the system.

6. Maintenance Required

- The application requires periodic updates, workflow maintenance, and security reviews to ensure smooth operation.

7. Role Management

- Incorrect ACL or role configuration can prevent users from accessing required information or performing authorized actions.

8. Limited Offline Access

- The application cannot be used without access to the ServiceNow instance.

9. FUTURE SCOPE

10.

The Employee Management System – Onboarding & Offboarding Automation developed on the ServiceNow platform has significant future enhancement opportunities. The current system automates employee onboarding, offboarding, approvals, notifications, and task management. In the future, the project can be expanded into a complete HR management solution with advanced features.

Future Enhancements

- Leave Management Integration
 - Add employee leave request and approval workflows.
 - Track leave balances and attendance records.
- Payroll Integration
 - Integrate the system with payroll applications.
 - Automatically update employee status changes for salary processing.
- Asset Management
 - Track laptops, ID cards, software licenses, and other assets assigned to employees.
 - Automate asset allocation during onboarding and asset return during offboarding.
- Employee Self-Service Portal
 - Allow employees to submit requests, check status, update profiles, and view notifications through a self-service portal.
- Advanced Reporting and Dashboards
 - Create real-time dashboards for HR managers.
 - Generate reports for onboarding trends, offboarding statistics, approval times, and departmental performance.
- AI-Based Automation
 - Use ServiceNow AI capabilities to predict delays, recommend actions, and automate repetitive HR tasks.
- Integration with External Systems
 - Connect with Active Directory, email systems, payroll software, and third-party HR applications.
- Mobile Application Support
 - Provide mobile access for managers and employees to approve requests and track activities from anywhere.
- Document Management
 - Store employee documents such as offer letters, ID proofs, and exit documents securely within the system.
- Multi-Department Workflow Expansion
 - Extend the solution to Finance, Administration, Facilities, and other departments for organization-wide process automation.

Conclusion

The future scope of this project is to transform the current onboarding and offboarding automation system into a comprehensive enterprise-level Human Resource Management solution. By integrating additional HR modules, analytics, AI capabilities, mobile access, and external systems, the application can provide a highly scalable, secure, and intelligent employee lifecycle management platform for organizations of any size.

GITHUB LINK:

<https://github.com/PepakayalaSaiGovind/automate-onboard-and-offboard-system.git>



Project Demonstration.mp4